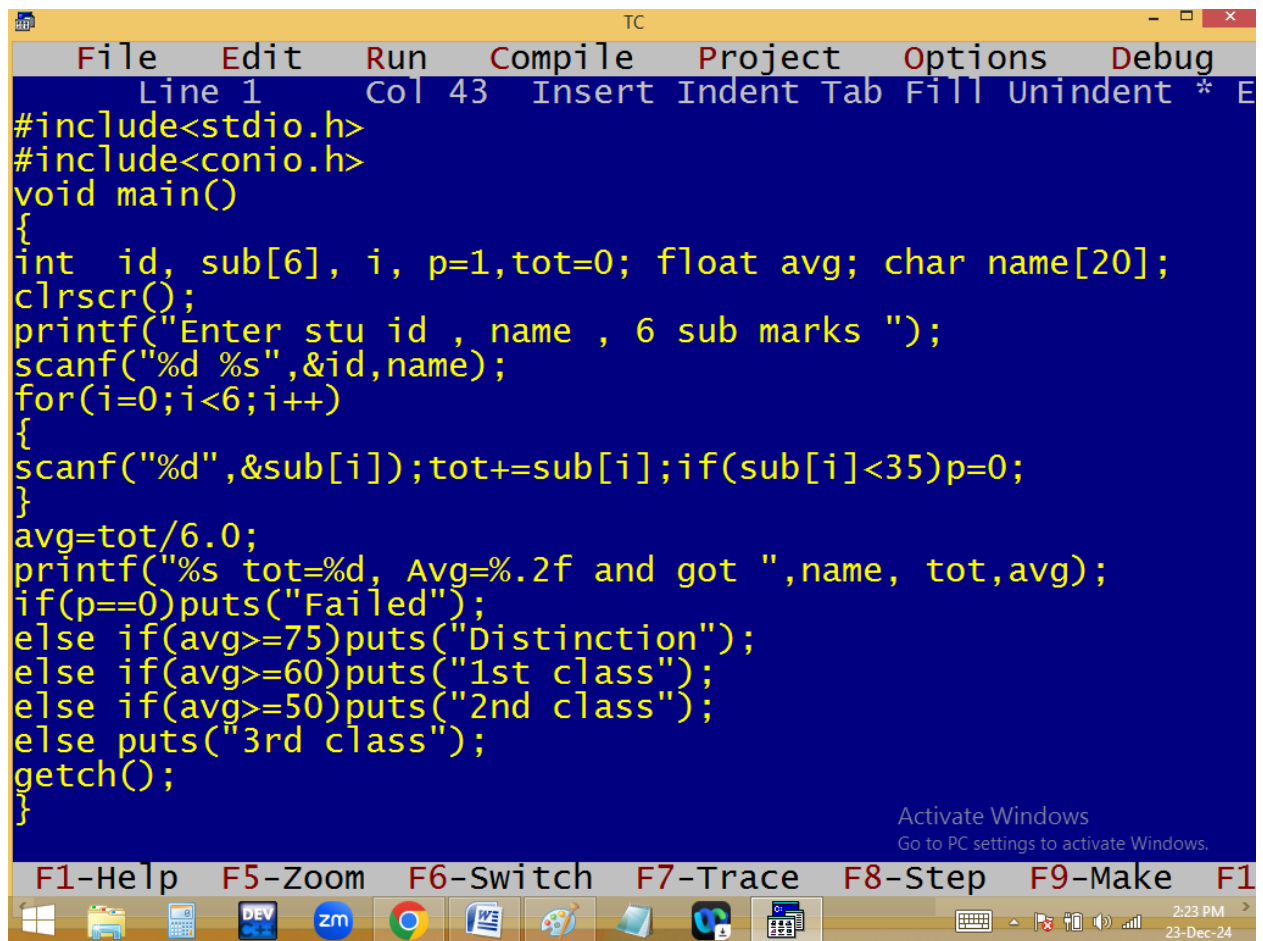


Finding student result using array:



The image shows a screenshot of the Turbo C++ (TC) IDE. The window title is "TC". The menu bar includes File, Edit, Run, Compile, Project, Options, and Debug. The status bar at the top indicates "Line 1 Col 43 Insert Indent Tab Fill Unindent * E". The code is written in yellow text on a blue background. It includes `<stdio.h>` and `<conio.h>`. The `main` function declares variables `id`, `sub[6]`, `i`, `p=1`, `tot=0`, `float avg`, and `char name[20]`. It uses `clrscr()` to clear the screen, then prompts the user to enter student ID and name. A loop calculates the total of six subject marks. After the loop, it calculates the average and prints the student's name, total marks, and average. It then uses a series of `if-else` statements to determine the student's grade based on the average. The program ends with `getch()`. The Windows taskbar at the bottom shows the Start button, several application icons (including DEV, zm, Chrome, Word, and a game controller), and the system tray with the time 2:23 PM and date 23-Dec-24. An "Activate Windows" watermark is visible in the bottom right corner of the IDE window.

```
#include<stdio.h>
#include<conio.h>
void main()
{
int id, sub[6], i, p=1,tot=0; float avg; char name[20];
clrscr();
printf("Enter stu id , name , 6 sub marks ");
scanf("%d %s",&id,name);
for(i=0;i<6;i++)
{
scanf("%d",&sub[i]);tot+=sub[i];if(sub[i]<35)p=0;
}
avg=tot/6.0;
printf("%s tot=%d, Avg=%.2f and got ",name, tot,avg);
if(p==0)puts("Failed");
else if(avg>=75)puts("Distinction");
else if(avg>=60)puts("1st class");
else if(avg>=50)puts("2nd class");
else puts("3rd class");
getch();
}
```

Activate Windows
Go to PC settings to activate Windows.

F1-Help F5-Zoom F6-Switch F7-Trace F8-Step F9-Make F10-Run

2:23 PM
23-Dec-24

```
TC
Enter stu id , name , 6 sub marks
1 krish 99 88 89 98 80 90
krish tot=544, Avg=90.67, and got Distinction
```

Activate Windows
Go to PC settings to activate Windows.

2:23 PM
23-Dec-24

```
TC
Enter stu id , name , 6 sub marks
1 krish 99 88 89 98 80 90
krish tot=544, Avg=90.67, and got Distinction
_
```

Activate Windows
Go to PC settings to activate Windows.

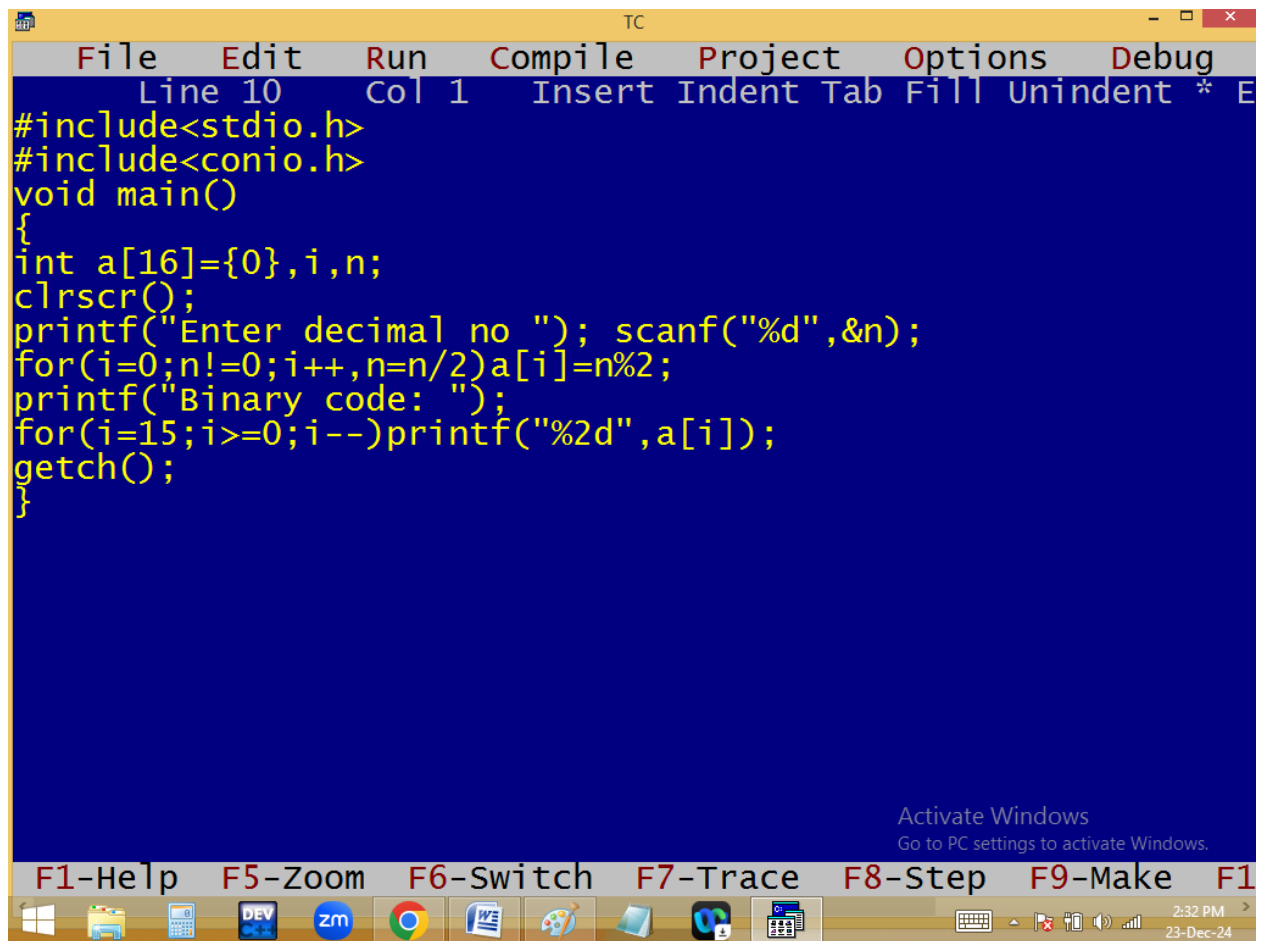
2:23 PM
23-Dec-24

```
TC
Enter stu id , name , 6 sub marks
2 bablu 54 45 43 30 55 45
bablu tot=272, Avg=45.33 and got Failed
```

Activate Windows
Go to PC settings to activate Windows.

Decimal to binary conversion:

20 → 0000 0000 0001 0100

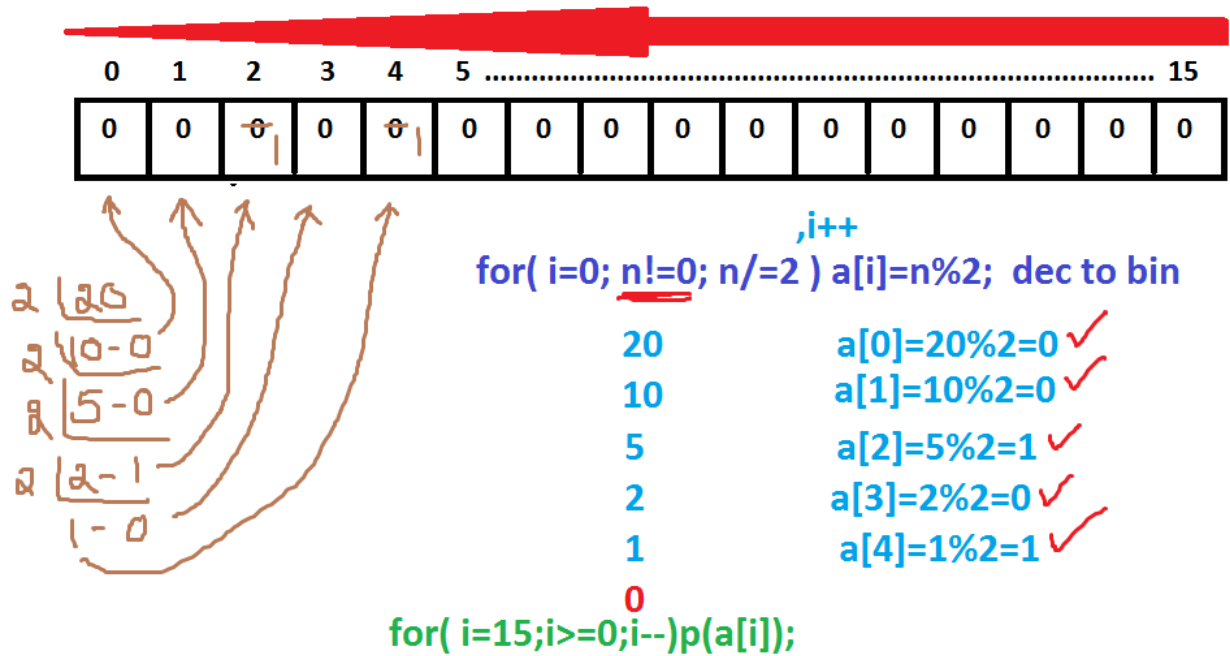
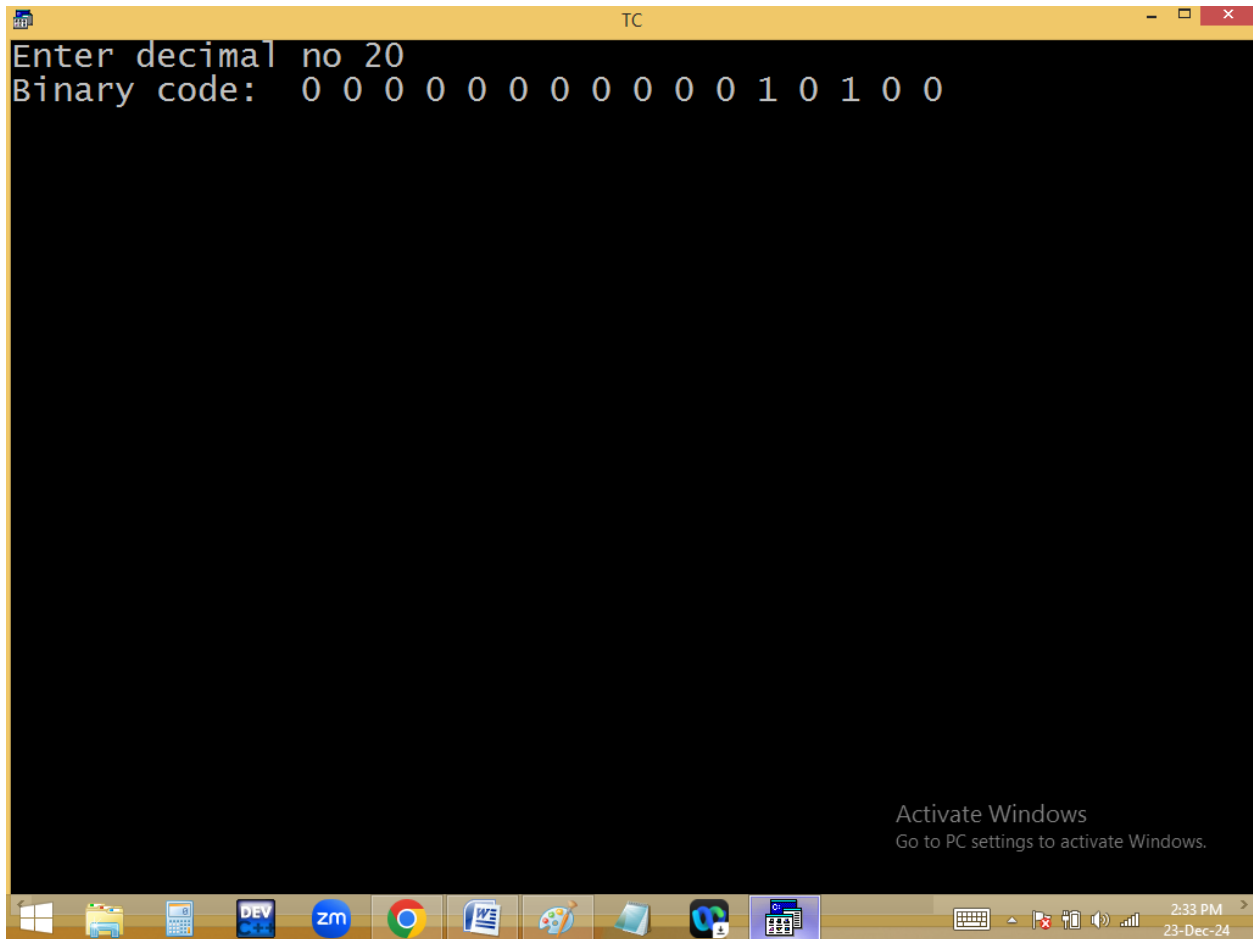


```
TC
File Edit Run Compile Project Options Debug
Line 10 Col 1 Insert Indent Tab Fill Unindent * E
#include<stdio.h>
#include<conio.h>
void main()
{
int a[16]={0},i,n;
clrscr();
printf("Enter decimal no "); scanf("%d",&n);
for(i=0;n!=0;i++,n=n/2)a[i]=n%2;
printf("Binary code: ");
for(i=15;i>=0;i--)printf("%2d",a[i]);
getch();
}
```

Activate Windows
Go to PC settings to activate Windows.

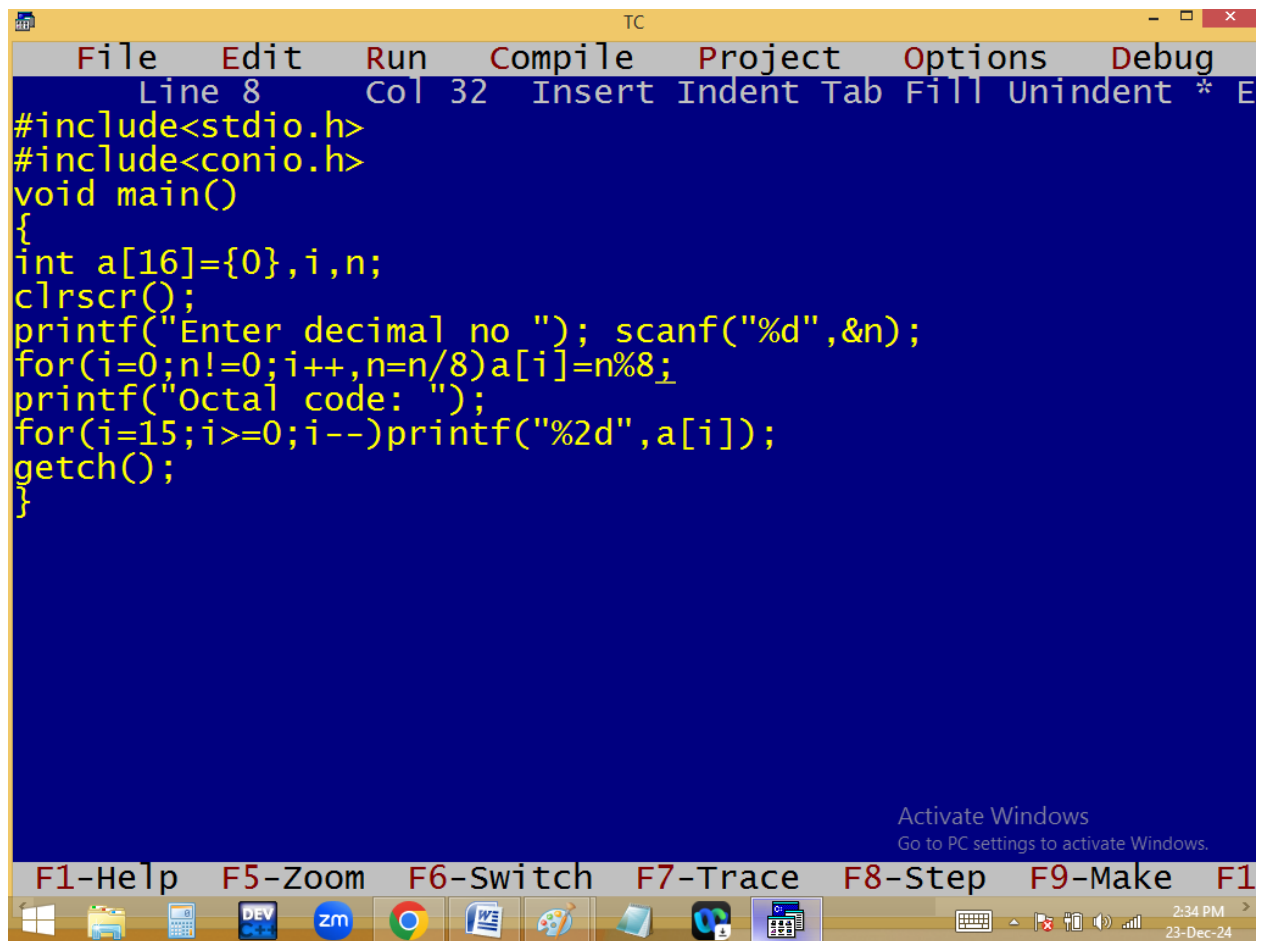
F1-Help F5-Zoom F6-Switch F7-Trace F8-Step F9-Make F10-Exit

2:32 PM
23-Dec-24



$$\begin{array}{r} 8 \overline{) 20} \\ 24 \end{array} \checkmark$$

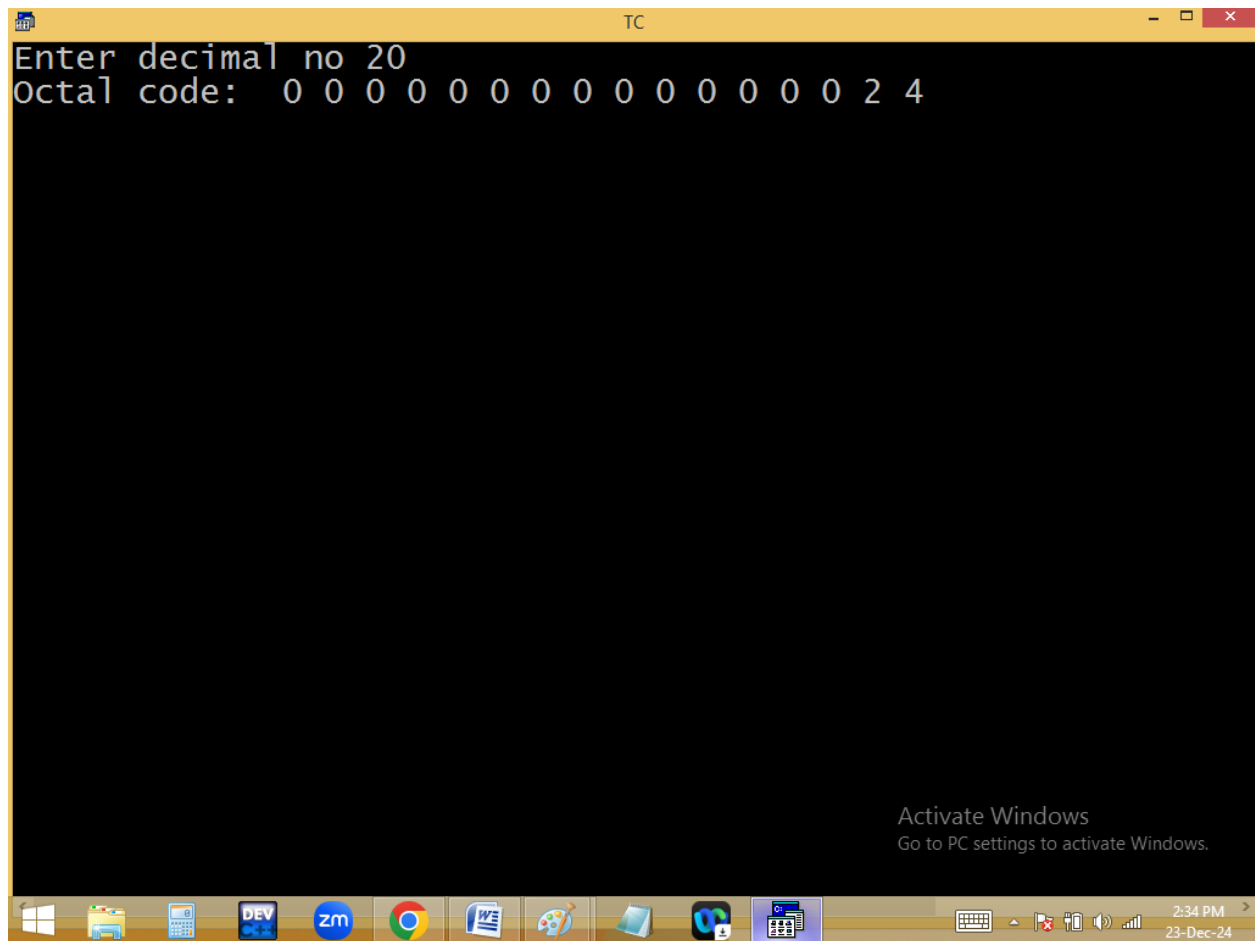
Decimal to octal:



The image shows a screenshot of the Turbo C++ (TC) IDE. The window title is "TC". The menu bar includes File, Edit, Run, Compile, Project, Options, and Debug. The status bar at the top indicates "Line 8 Col 32" and provides shortcuts for Insert, Indent, Tab, Fill, Unindent, and End of Line (* E). The code editor contains the following C program:

```
#include<stdio.h>
#include<conio.h>
void main()
{
int a[16]={0},i,n;
clrscr();
printf("Enter decimal no "); scanf("%d",&n);
for(i=0;n!=0;i++,n=n/8)a[i]=n%8;
printf("Octal code: ");
for(i=15;i>=0;i--)printf("%2d",a[i]);
getch();
}
```

At the bottom of the IDE, there is a row of function key shortcuts: F1-Help, F5-Zoom, F6-Switch, F7-Trace, F8-Step, F9-Make, and F10-Run. Below the IDE window is the Windows taskbar, which includes icons for the Start menu, File Explorer, Calculator, DEV C++, Zoom, Google Chrome, Word, Paint, and a folder. The system tray on the right shows the time as 2:34 PM and the date as 23-Dec-24. An "Activate Windows" watermark is visible in the bottom right corner of the desktop area.



The image shows a screenshot of a Turbo C++ (TC) window. The window title bar is yellow and contains the text "TC". The main area is black with white text. The text reads: "Enter decimal no 20" followed by "Octal code: 0 0 0 0 0 0 0 0 0 0 0 0 0 0 2 4". The window is running on a Windows desktop. The taskbar at the bottom shows various icons including the Start button, File Explorer, Calculator, DEV C++, Zm, Chrome, Word, Paint, and a folder. The system tray on the right shows the time as 2:34 PM and the date as 23-Dec-24. A watermark "Activate Windows" is visible in the bottom right corner of the window.

```
Enter decimal no 20
Octal code: 0 0 0 0 0 0 0 0 0 0 0 0 0 0 2 4
```

Decimal to hexadecimal:

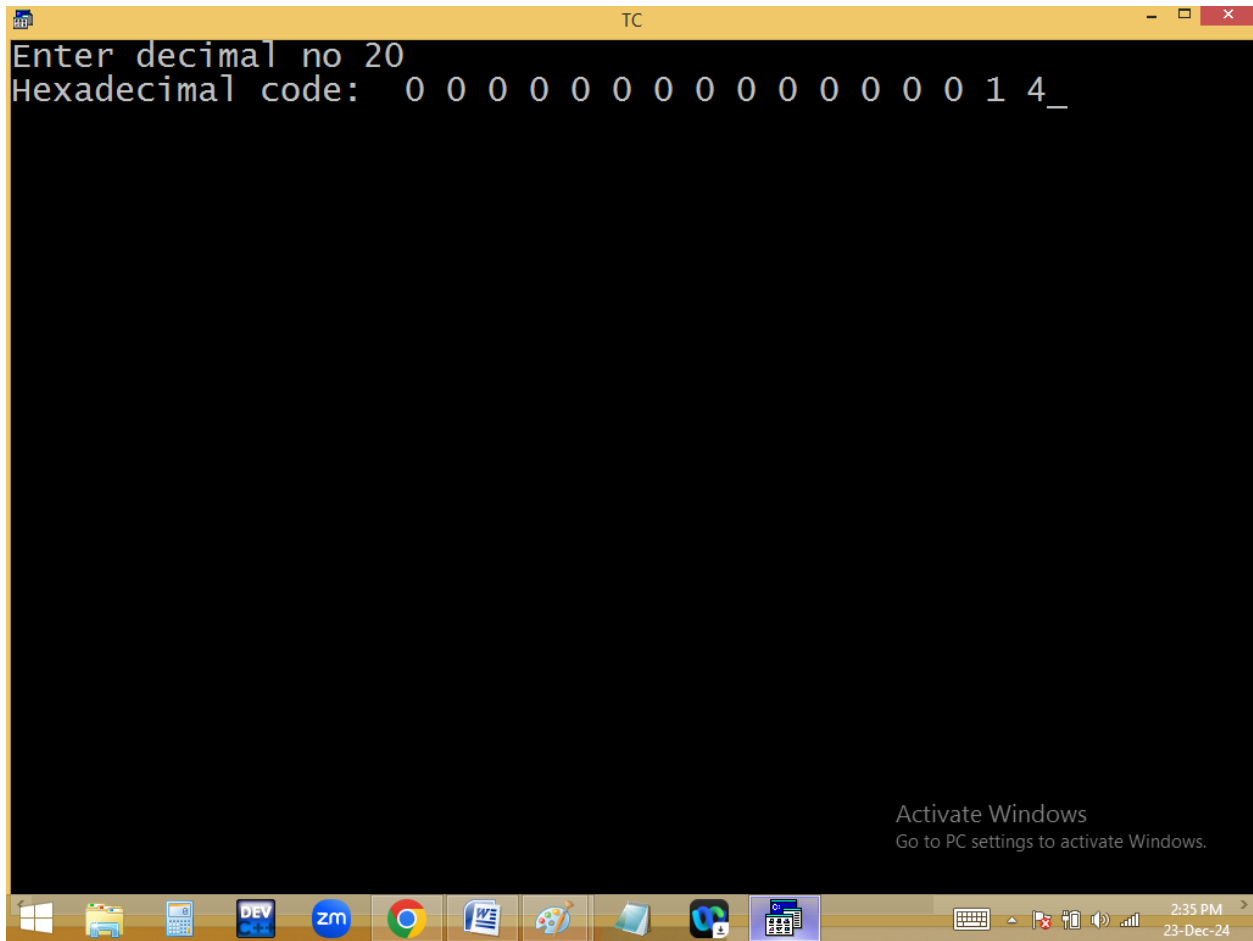
$$\begin{array}{r} 16 \overline{) 20} \\ \underline{16} \\ 4 \end{array}$$

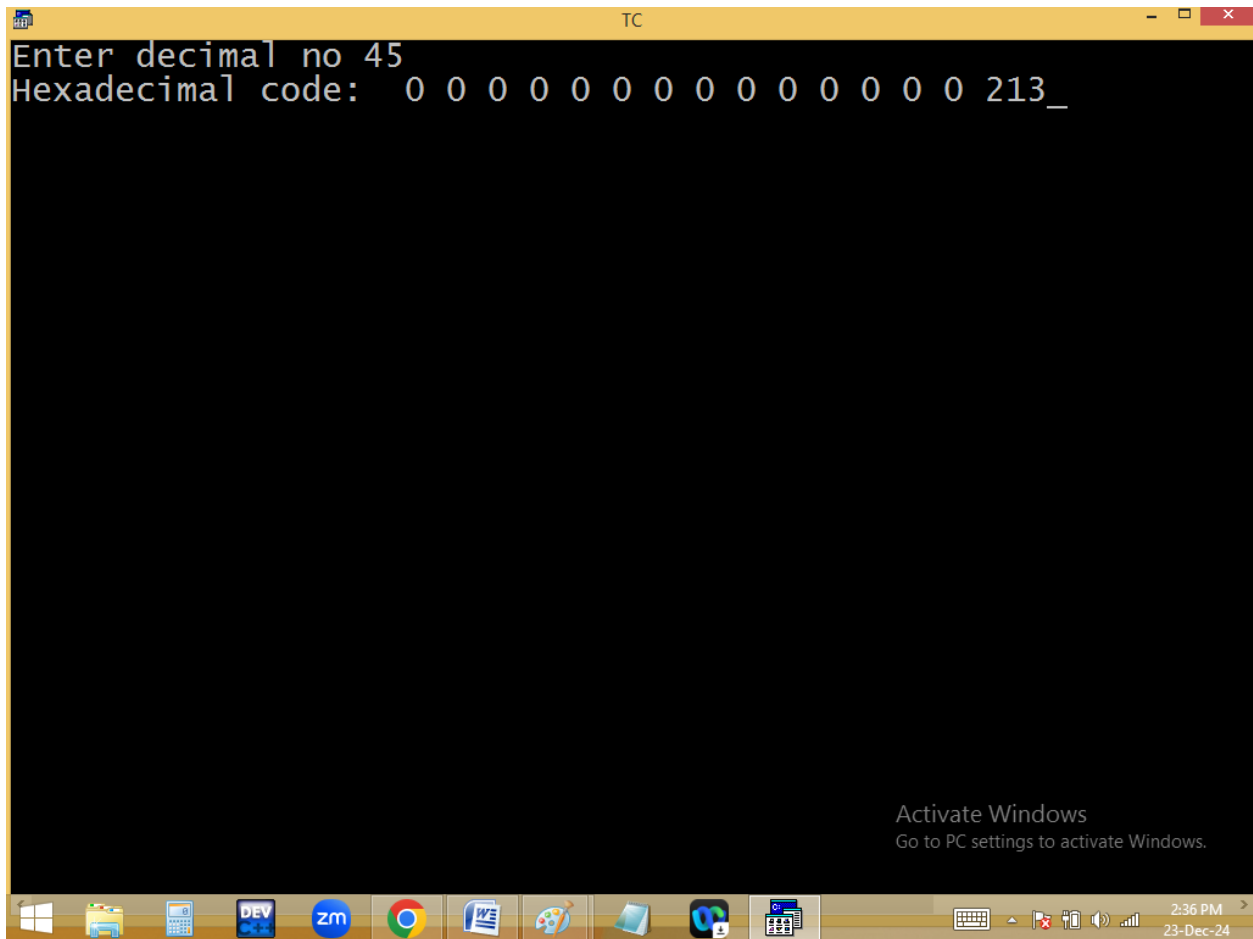
```
TC
File Edit Run Compile Project Options Debug
Line 9 Col 20 Insert Indent Tab Fill Unindent * E
#include<stdio.h>
#include<conio.h>
void main()
{
int a[16]={0},i,n;
clrscr();
printf("Enter decimal no "); scanf("%d",&n);
for(i=0;n!=0;i++,n=n/16)a[i]=n%16;
printf("Hexadecimal code: ");
for(i=15;i>=0;i--)printf("%2d",a[i]);
getch();
}
```

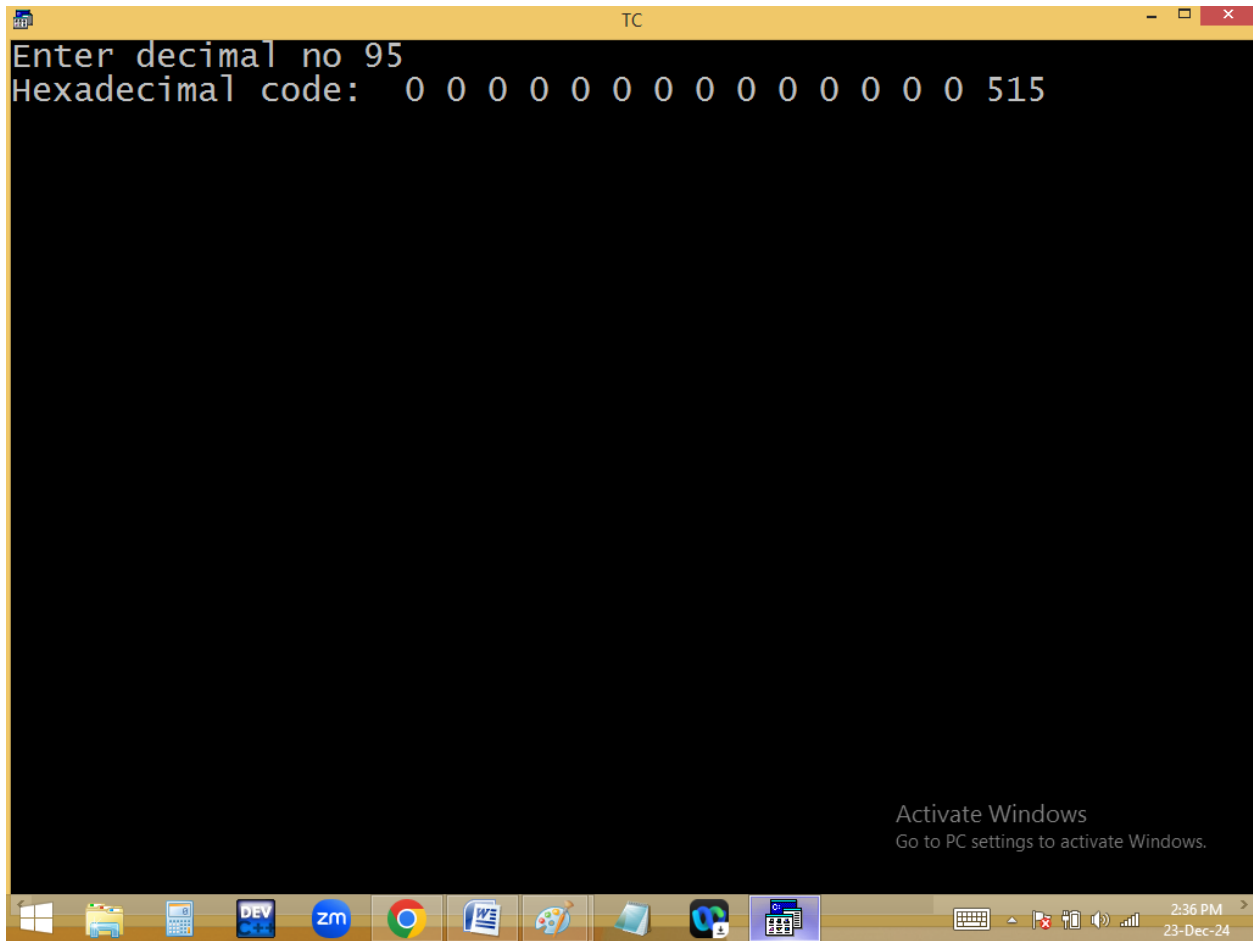
Activate Windows
Go to PC settings to activate Windows.

F1-Help F5-Zoom F6-Switch F7-Trace F8-Step F9-Make F10-Run

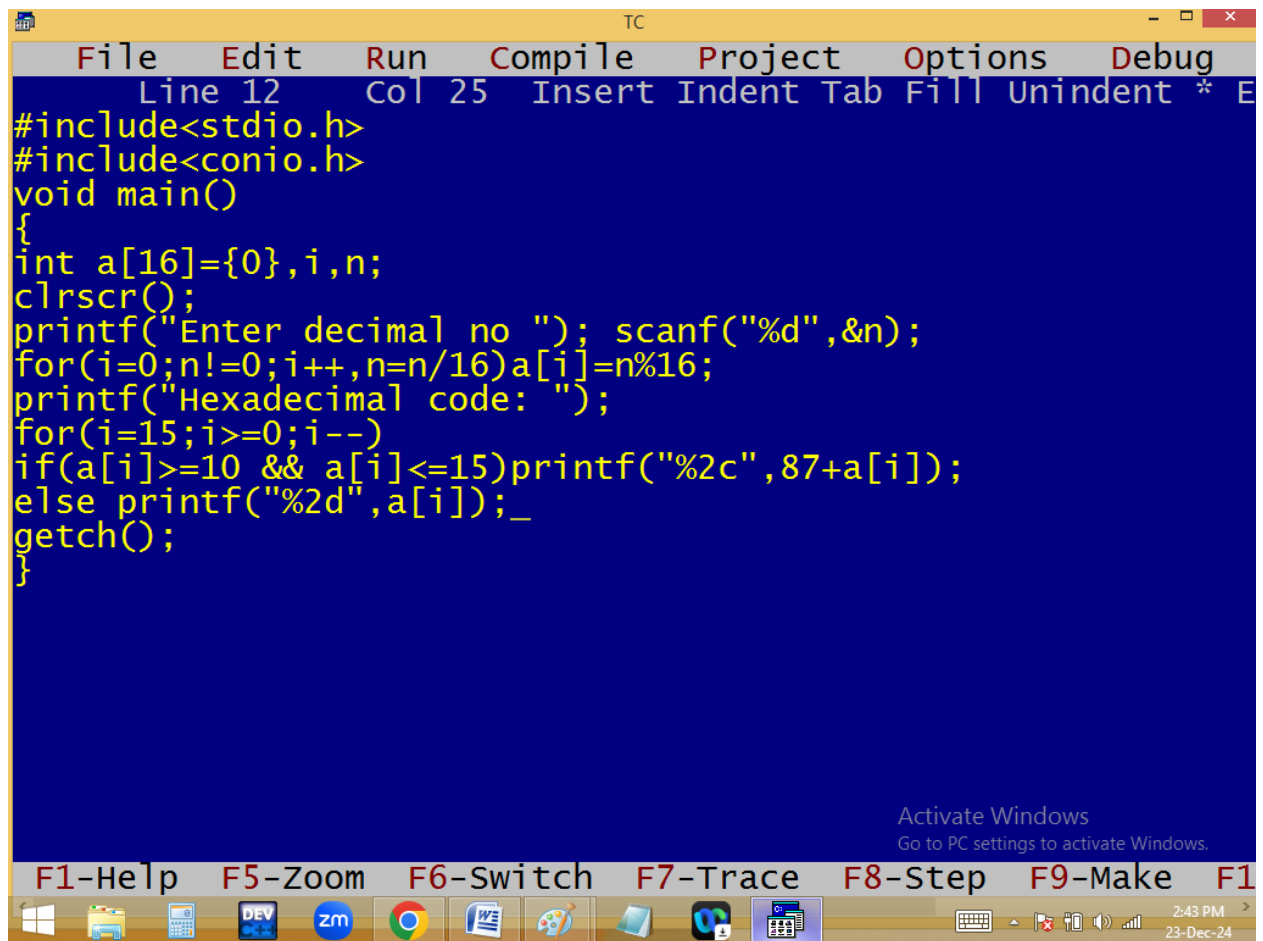
2:35 PM
23-Dec-24







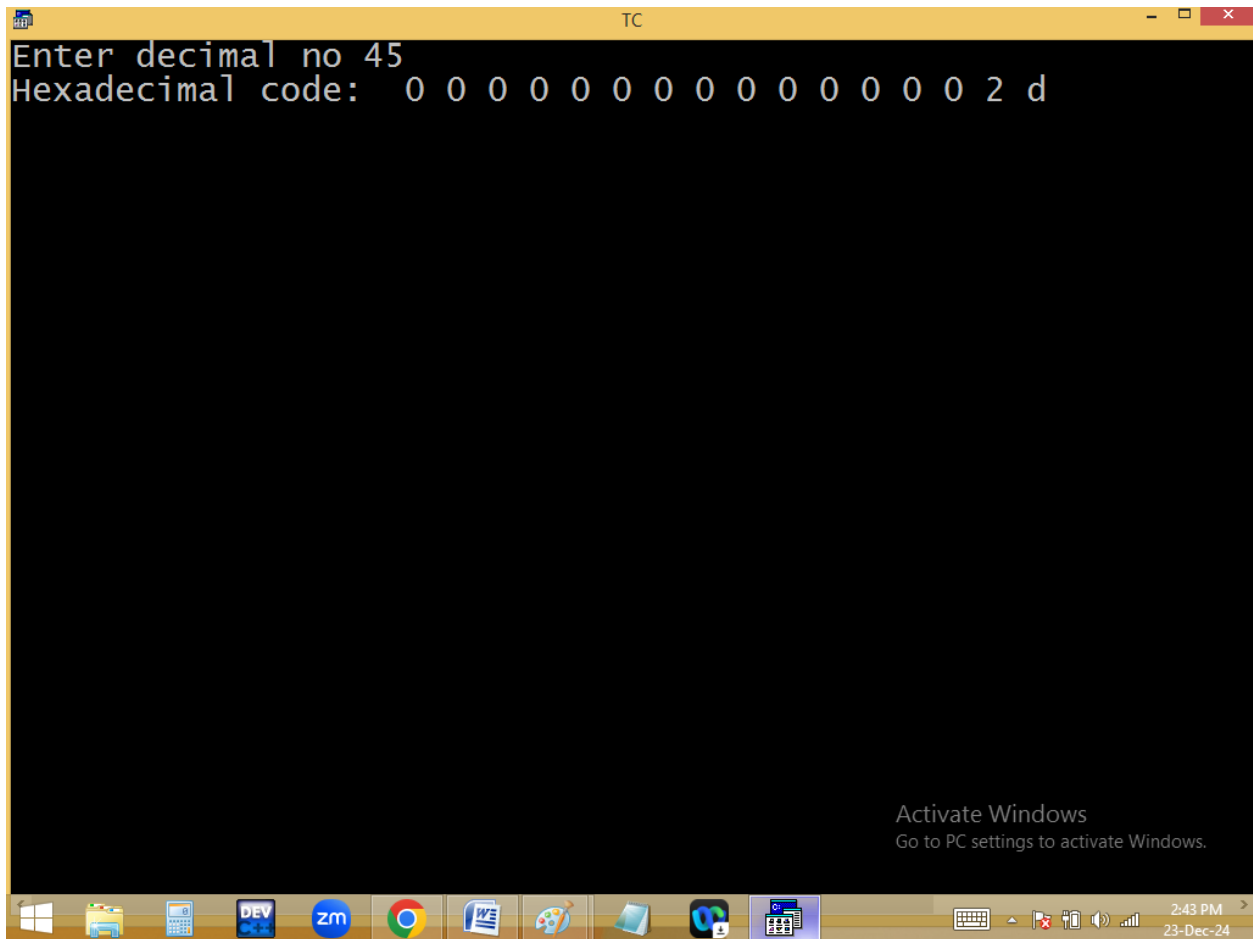
```
Enter decimal no 95  
Hexadecimal code: 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 515
```

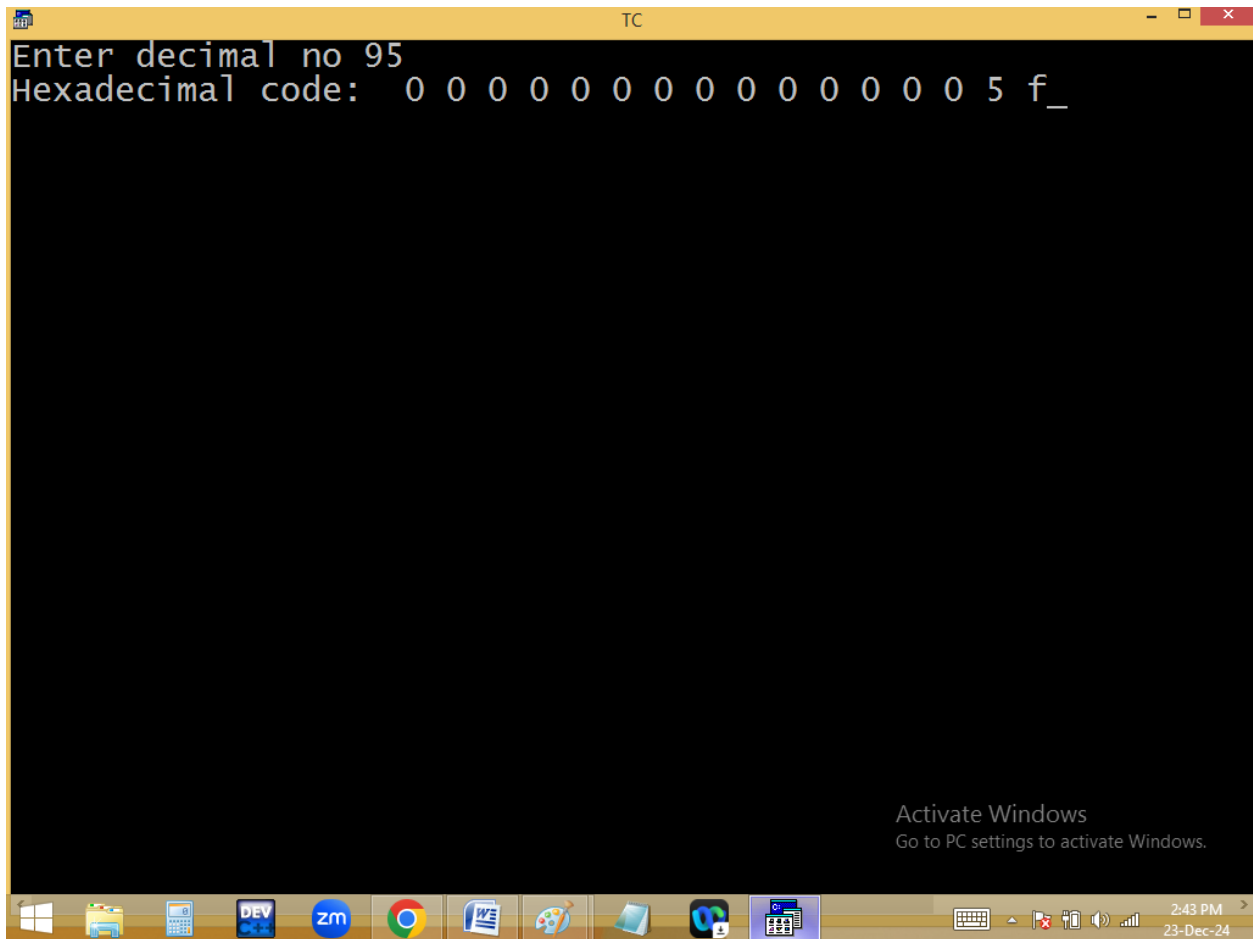


The image shows a screenshot of a Turbo C++ (TC) IDE window. The window has a yellow title bar with the text "TC" and standard window controls. Below the title bar is a menu bar with the following options: File, Edit, Run, Compile, Project, Options, and Debug. Under the "Run" menu, there are sub-options: Line 12, Col 25, Insert, Indent, Tab, Fill, Unindent, and a symbol. The main editing area has a dark blue background with yellow text. The code is as follows:

```
#include<stdio.h>
#include<conio.h>
void main()
{
int a[16]={0},i,n;
clrscr();
printf("Enter decimal no "); scanf("%d",&n);
for(i=0;n!=0;i++,n=n/16)a[i]=n%16;
printf("Hexadecimal code: ");
for(i=15;i>=0;i--)
if(a[i]>=10 && a[i]<=15)printf("%2c",87+a[i]);
else printf("%2d",a[i]);_
getch();
}
```

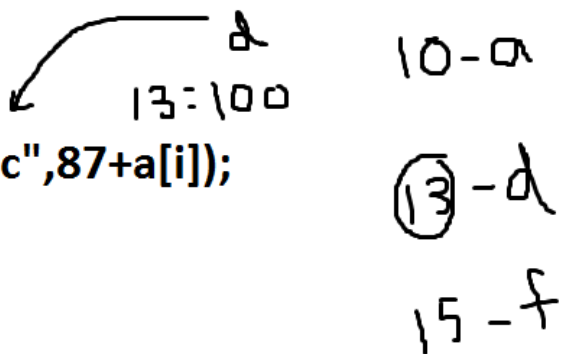
At the bottom of the window, there is a status bar with the text "Activate Windows" and "Go to PC settings to activate Windows." Below the status bar is a taskbar with several icons: Windows Start button, File Explorer, Calculator, DEV C++, Zoom, Google Chrome, Microsoft Word, Paint, and a folder icon. To the right of the taskbar, the system clock shows "2:43 PM" and the date "23-Dec-24".





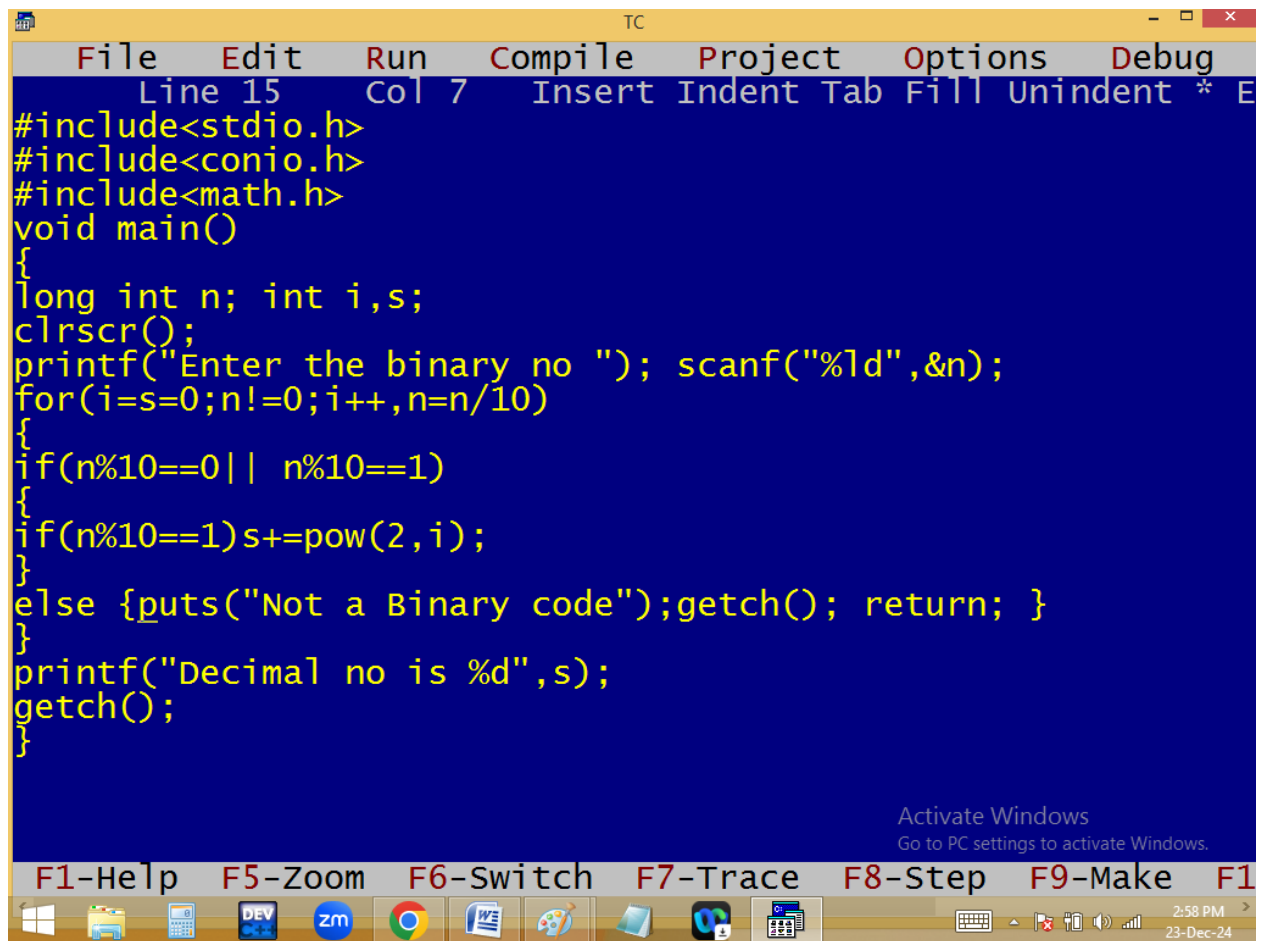
```
TC
Enter decimal no 20
Hexadecimal code: 0 0 0 0 0 0 0 0 0 0 0 0 0 0 1 4_

Activate Windows
Go to PC settings to activate Windows.
```



```
if(a[i]>=10 && a[i]<=15) p("%2c",87+a[i]);
else p(a[i]);
```

Binary to decimal:



The image shows a screenshot of a Turbo C++ (TC) IDE window. The window has a yellow title bar with the text "TC" and standard window controls. The menu bar includes "File", "Edit", "Run", "Compile", "Project", "Options", and "Debug". Below the menu bar, a status bar shows "Line 15", "Col 7", and various editing options like "Insert", "Indent", "Tab", "Fill", "Unindent", and a comment toggle. The main editing area has a blue background and contains the following C code:

```
#include<stdio.h>
#include<conio.h>
#include<math.h>
void main()
{
    long int n; int i,s;
    clrscr();
    printf("Enter the binary no "); scanf("%ld",&n);
    for(i=s=0;n!=0;i++,n=n/10)
    {
        if(n%10==0 || n%10==1)
        {
            if(n%10==1)s+=pow(2,i);
        }
        else {puts("Not a Binary code");getch(); return; }
    }
    printf("Decimal no is %d",s);
    getch();
}
```

At the bottom of the window, there is a taskbar with several icons including Windows Explorer, Calculator, DEV C++, Zoom, Google Chrome, Word, Paint, and a folder. To the right of the taskbar, there is a system tray showing the time "2:58 PM" and the date "23-Dec-24". An "Activate Windows" watermark is visible in the bottom right corner of the IDE window.

Line 15 Col 7 Insert Indent Tab Fill Unindent * E

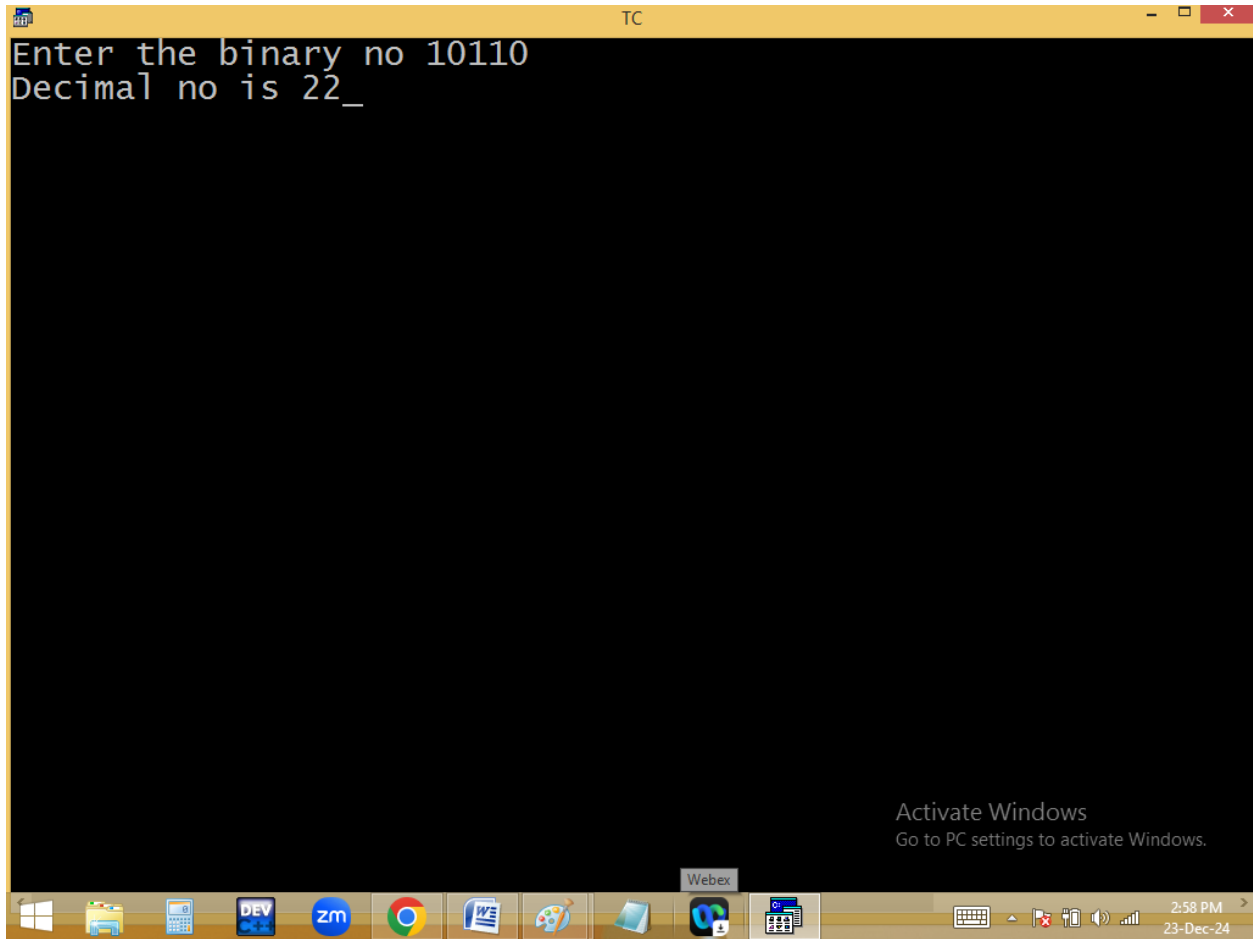
File Edit Run Compile Project Options Debug

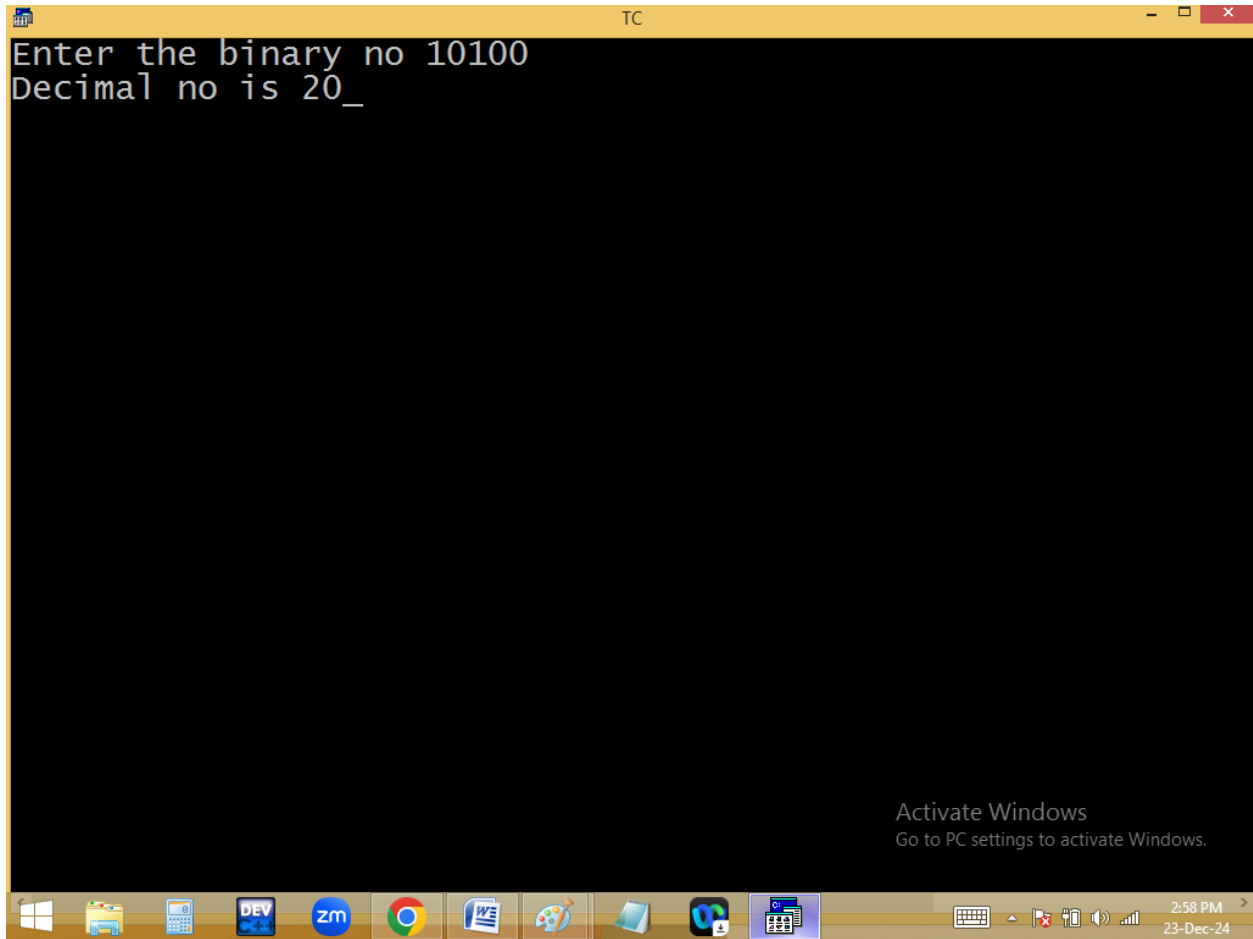
#include<stdio.h>
#include<conio.h>
#include<math.h>
void main()
{
 long int n; int i,s;
 clrscr();
 printf("Enter the binary no "); scanf("%ld",&n);
 for(i=s=0;n!=0;i++,n=n/10)
 {
 if(n%10==0 || n%10==1)
 {
 if(n%10==1)s+=pow(2,i);
 }
 else {puts("Not a Binary code");getch(); return; }
 }
 printf("Decimal no is %d",s);
 getch();
}

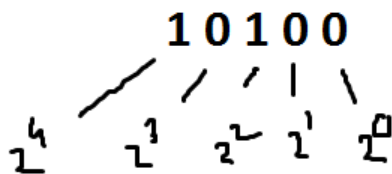
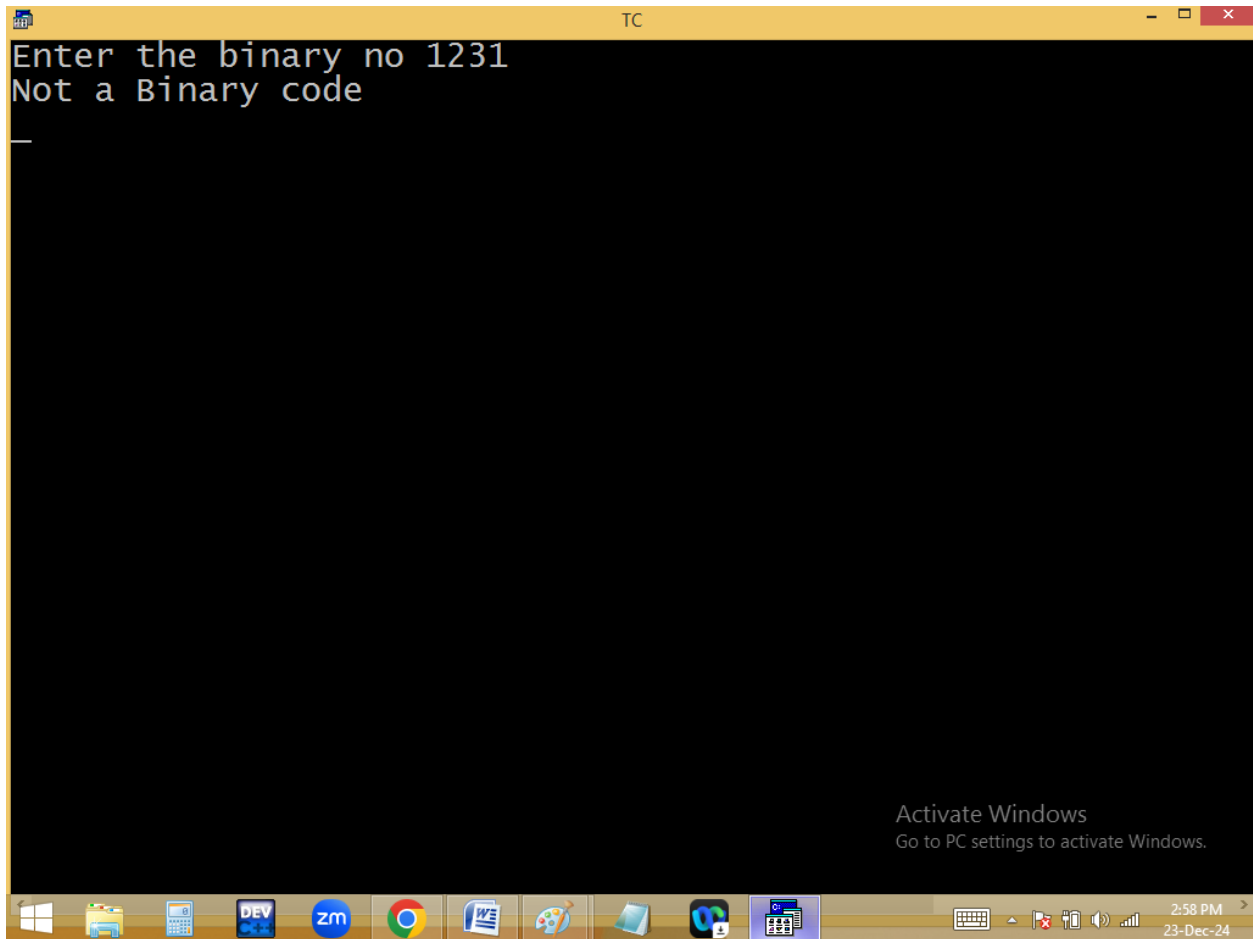
Activate Windows
Go to PC settings to activate Windows.

F1-Help F5-Zoom F6-Switch F7-Trace F8-Step F9-Make F10-Run

2:58 PM
23-Dec-24



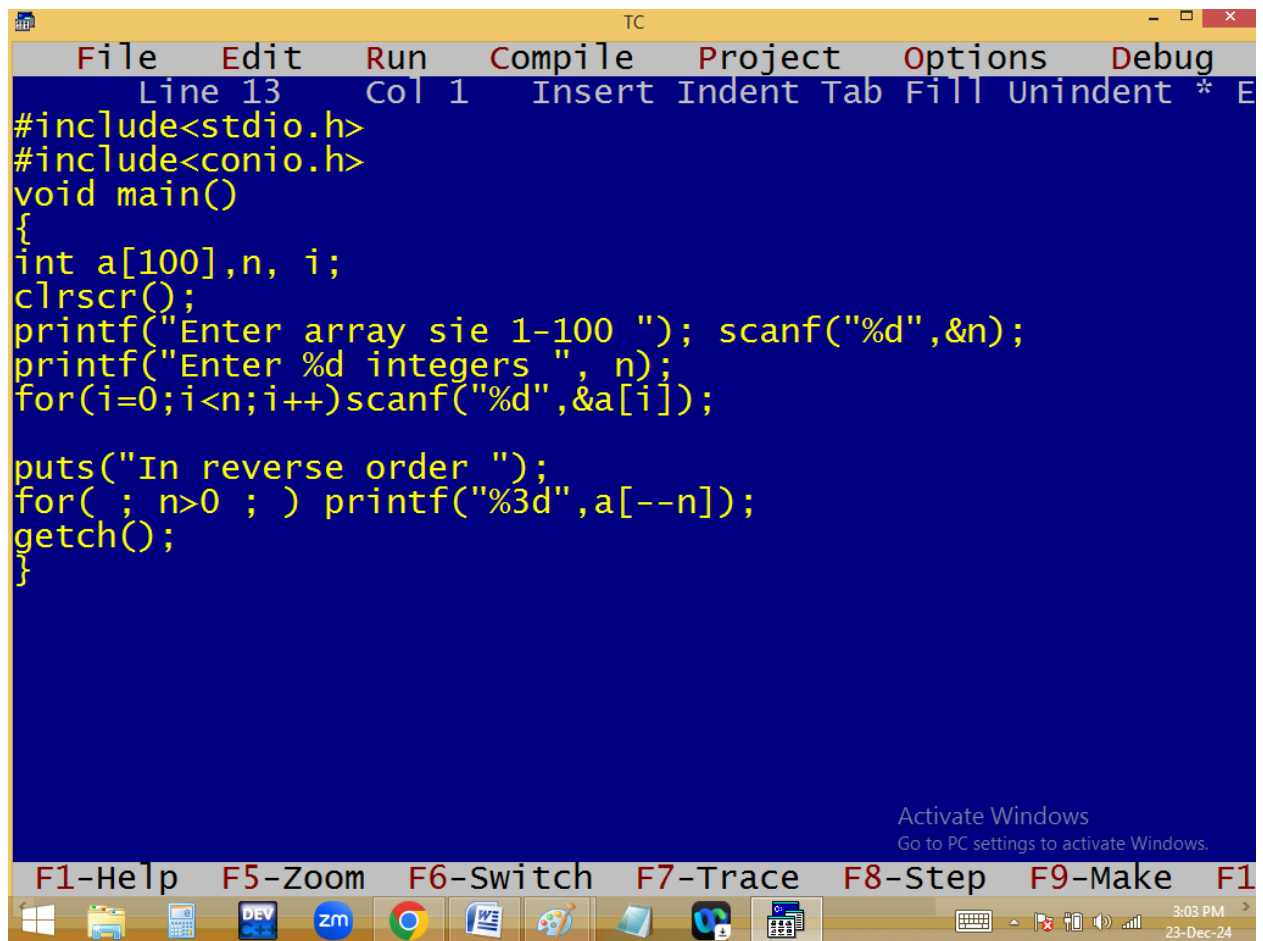




```
for( s=i=0; n!=0;i++,n=n/10)
{
  if(n%10==1)s=s+pow(2,i);
}
p(s);
```

Arranging array elements in reverse order:

Temp method:



The image shows a screenshot of a Turbo C++ (TC) IDE window. The window has a yellow title bar with the text "TC" and standard window controls. Below the title bar is a menu bar with the following options: File, Edit, Run, Compile, Project, Options, and Debug. Under the "Run" menu, there is a submenu with the following options: Line 13, Col 1, Insert, Indent, Tab, Fill, Unindent, *, and E. The main editing area has a blue background and contains the following C code:

```
#include<stdio.h>
#include<conio.h>
void main()
{
int a[100],n, i;
clrscr();
printf("Enter array size 1-100 "); scanf("%d",&n);
printf("Enter %d integers ", n);
for(i=0;i<n;i++)scanf("%d",&a[i]);

puts("In reverse order ");
for( ; n>0 ; ) printf("%3d",a[--n]);
getch();
}
```

At the bottom of the window, there is a status bar with the text "Activate Windows" and "Go to PC settings to activate Windows." Below the status bar is a taskbar with the following icons: Windows Start button, File Explorer, Calculator, DEV C++, Zoom, Google Chrome, Microsoft Word, Paint, and a folder icon. The taskbar also shows the system clock as 3:03 PM on 23-Dec-24.

```
TC
Enter array size 1-100 5
Enter 5 integers 1 0 1 0 0
In reverse order
0 0 1 0 1
```

Activate Windows
Go to PC settings to activate Windows.

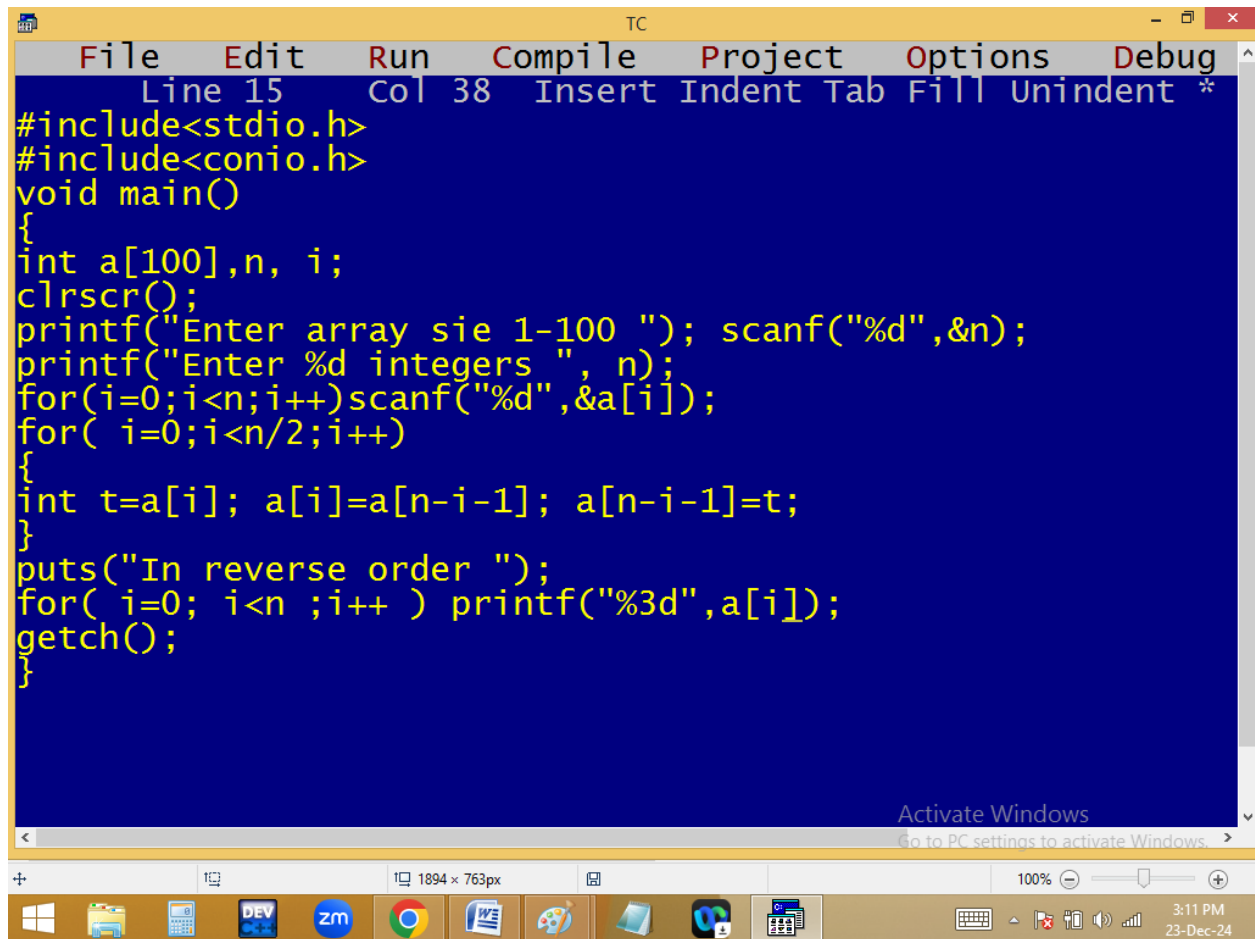
3:03 PM
23-Dec-24

0	1	2	3	4
0	0	1	0	1

$\frac{n}{5}$

```
p("Reverse order");  
for( ; n > 0; ) p(a[--n]);
```

Permanent arrangement:



The image shows a screenshot of a Turbo C++ (TC) IDE window. The window has a yellow title bar with the text "TC" and standard window controls. Below the title bar is a menu bar with the following options: File, Edit, Run, Compile, Project, Options, and Debug. A status bar at the top of the editor area shows "Line 15 Col 38" and a list of editing actions: Insert, Indent, Tab, Fill, Unindent, and *. The main editor area has a dark blue background with yellow text. The code is as follows:

```
#include<stdio.h>
#include<conio.h>
void main()
{
int a[100],n, i;
clrscr();
printf("Enter array size 1-100 "); scanf("%d",&n);
printf("Enter %d integers ", n);
for(i=0;i<n;i++)scanf("%d",&a[i]);
for( i=0;i<n/2;i++)
{
int t=a[i]; a[i]=a[n-i-1]; a[n-i-1]=t;
}
puts("In reverse order ");
for( i=0; i<n ;i++ ) printf("%3d",a[i]);
getch();
}
```

At the bottom of the window, there is a taskbar with various icons including Windows, File Explorer, Calculator, DEV C++, ZOOM, Google Chrome, Word, Paint, and a folder icon. The system tray on the right shows the time as 3:11 PM and the date as 23-Dec-24. A watermark "Activate Windows" is visible in the bottom right corner of the IDE window.

```
TC
Enter array size 1-100 5
Enter 5 integers 9 0 7 5 8
In reverse order
8 5 7 0 9
```

Activate Windows
Go to PC settings to activate Windows.

Page: 27 of 27 | Words: 34 | 110%

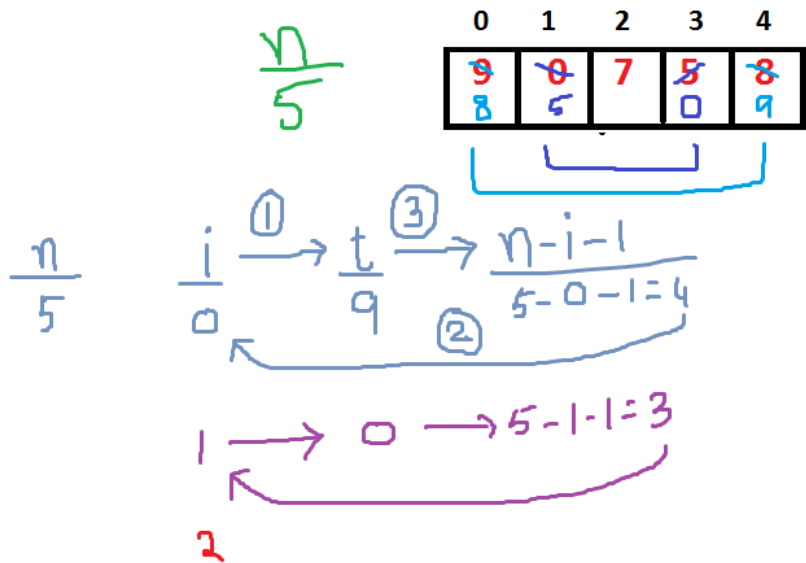
3:11 PM
23-Dec-24

```
TC
Enter array size 1-100 4
Enter 4 integers 1 0 2 7
In reverse order
7 2 0 1
```

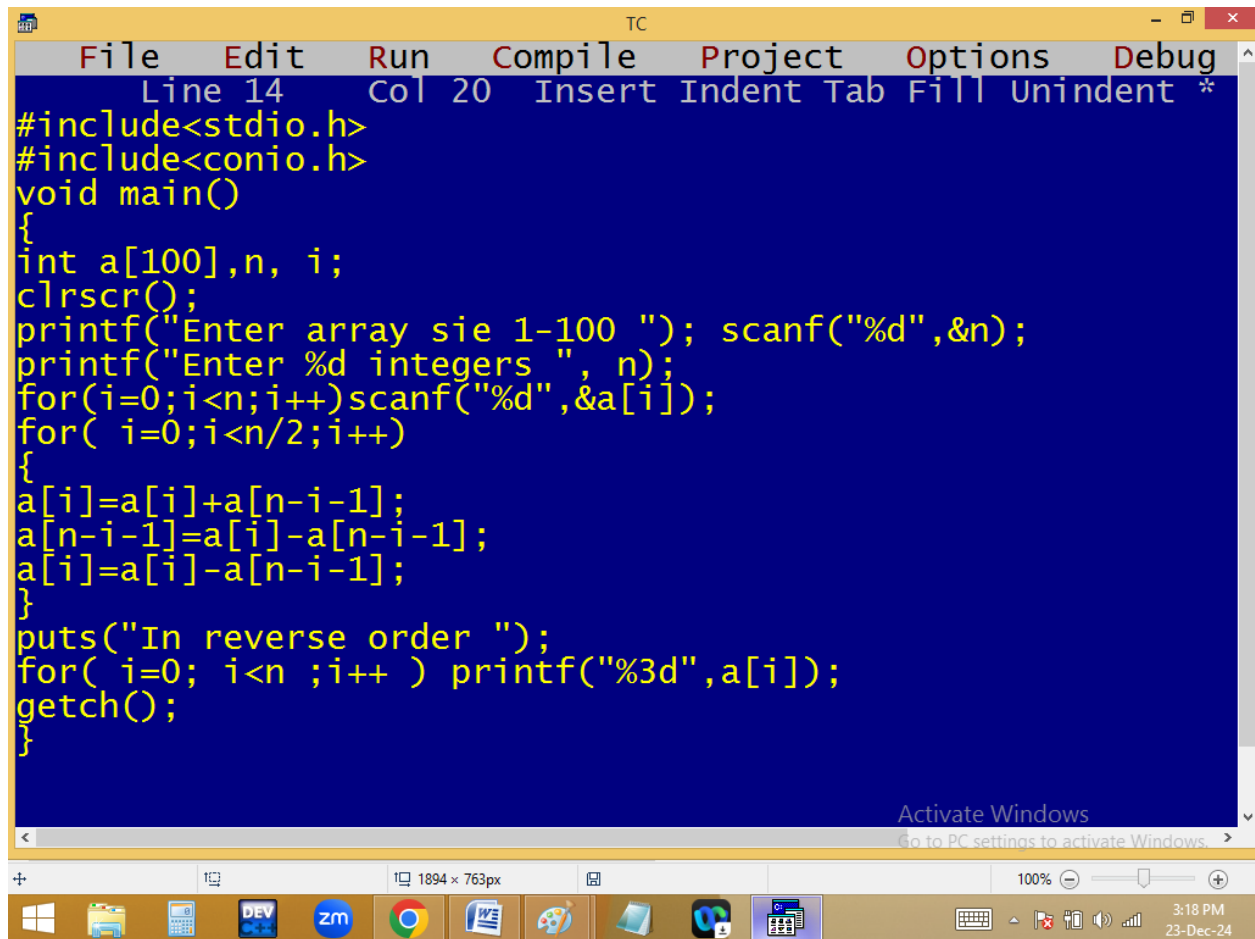
Activate Windows
Go to PC settings to activate Windows.

Page: 28 of 28 Words: 34 110% 3:12 PM 23-Dec-24

```
for( i=0; i<n/2; i++ )
{
    int t=a[i];
    a[i]=a[n-i-1];
    a[n-i-1]=t;
}
```



Without using temp variable:



The image shows a screenshot of a Turbo C++ (TC) IDE window. The title bar reads "TC". The menu bar includes "File", "Edit", "Run", "Compile", "Project", "Options", and "Debug". Below the menu bar, the status bar shows "Line 14", "Col 20", and "Insert Indent Tab Fill Unindent *". The main editing area has a blue background with yellow text. The code is as follows:

```
#include<stdio.h>
#include<conio.h>
void main()
{
int a[100],n, i;
clrscr();
printf("Enter array size 1-100 "); scanf("%d",&n);
printf("Enter %d integers ", n);
for(i=0;i<n;i++)scanf("%d",&a[i]);
for( i=0;i<n/2;i++)
{
a[i]=a[i]+a[n-i-1];
a[n-i-1]=a[i]-a[n-i-1];
a[i]=a[i]-a[n-i-1];
}
puts("In reverse order ");
for( i=0; i<n ;i++ ) printf("%3d",a[i]);
getch();
}
```

At the bottom of the window, there is a taskbar with various application icons (Windows, File Explorer, Calculator, DEV C++, ZOOM, Chrome, Word, Paint, VS Code, and a terminal). The system tray on the right shows the time as 3:18 PM and the date as 23-Dec-24. A watermark "Activate Windows" is visible in the bottom right corner of the IDE window.

```

Enter array size 1-100 7
Enter 7 integers 1 2 3 4 5 6 7
In reverse order
7 6 5 4 3 2 1

```

Activate Windows
Go to PC settings to activate Windows.

Page: 30 of 31 Words: 38

3:18 PM
23-Dec-24

Handwritten Notes:

$\frac{n}{5}$

$9+8=17-9=8$ $17-8=9$

0 1 2 3 4

9	8	7	5	8
8	5		0	9

$5-0=5$

$\frac{n}{5}$

$i \xrightarrow{(1)} t \xrightarrow{(3)} n-i-1$

$0 \xrightarrow{(2)} 9 \xrightarrow{(2)} 5-0-1=4$

$1 \rightarrow 0 \rightarrow 5-1-1=3$

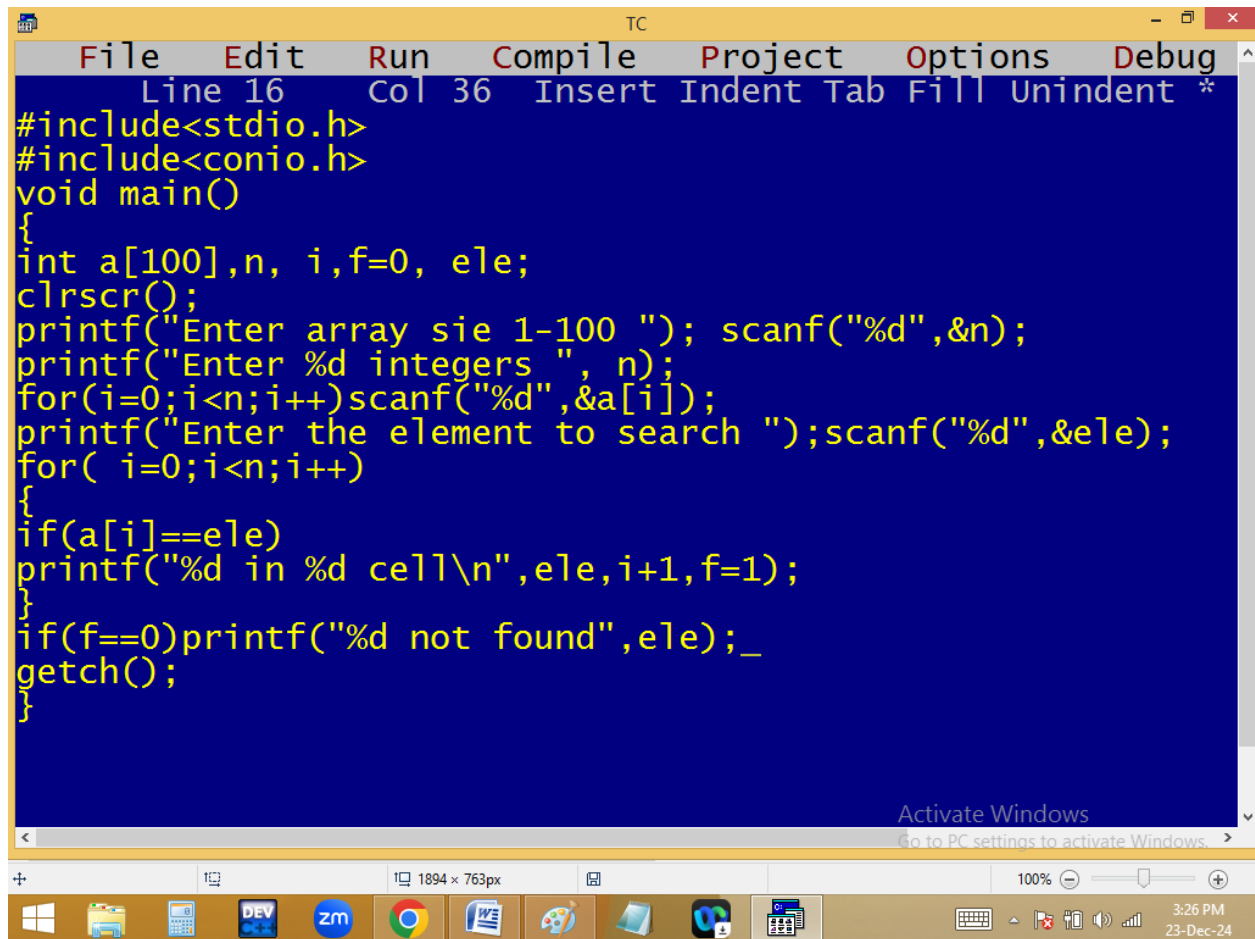
2

```

for( i=0; i<n/2; i++ )
{
    a[i]=a[i]+a[n-i-1];
    a[n-i-1]=a[i]-a[n-i-1];
    a[i]=a[i]-a[n-i-1];
}

```


Linear search:



The image shows a screenshot of the Turbo C++ (TC) IDE. The window title is "TC". The menu bar includes "File", "Edit", "Run", "Compile", "Project", "Options", and "Debug". The status bar at the top indicates "Line 16", "Col 36", and "Insert Indent Tab Fill Unindent *". The code is written in C and is as follows:

```
#include<stdio.h>
#include<conio.h>
void main()
{
int a[100],n, i,f=0, ele;
clrscr();
printf("Enter array size 1-100 "); scanf("%d",&n);
printf("Enter %d integers ", n);
for(i=0;i<n;i++)scanf("%d",&a[i]);
printf("Enter the element to search ");scanf("%d",&ele);
for( i=0;i<n;i++)
{
if(a[i]==ele)
printf("%d in %d cell\n",ele,i+1,f=1);
}
if(f==0)printf("%d not found",ele);_
getch();
}
```

At the bottom of the IDE window, there is a watermark that says "Activate Windows" and "Go to PC settings to activate Windows." The Windows taskbar is visible at the bottom, showing various application icons and the system clock indicating 3:26 PM on 23-Dec-24.

```
TC
Enter array size 1-100 9
Enter 9 integers 1 2 0 1 7 4 1 8 1
Enter the element to search 1
1 in 1 cell
1 in 4 cell
1 in 7 cell
1 in 9 cell
```

Activate Windows
Go to PC settings to activate Windows.

Page: 33 of 34 Words: 40 110%

3:26 PM
23-Dec-24

```

TC
Enter array size 1-100 5
Enter 5 integers 1 2 3 4 5
Enter the element to search 9
9 not found_

```

Activate Windows
Go to PC settings to activate Windows.

Page: 35 of 36 Words: 40 110% 3:26 PM 23-Dec-24

```

for( i=0; i<5; i++ )
{
    if( a[i] == ele )
    p("%d in %d cell\n",ele,i+1,f=1);
}

```

1 4

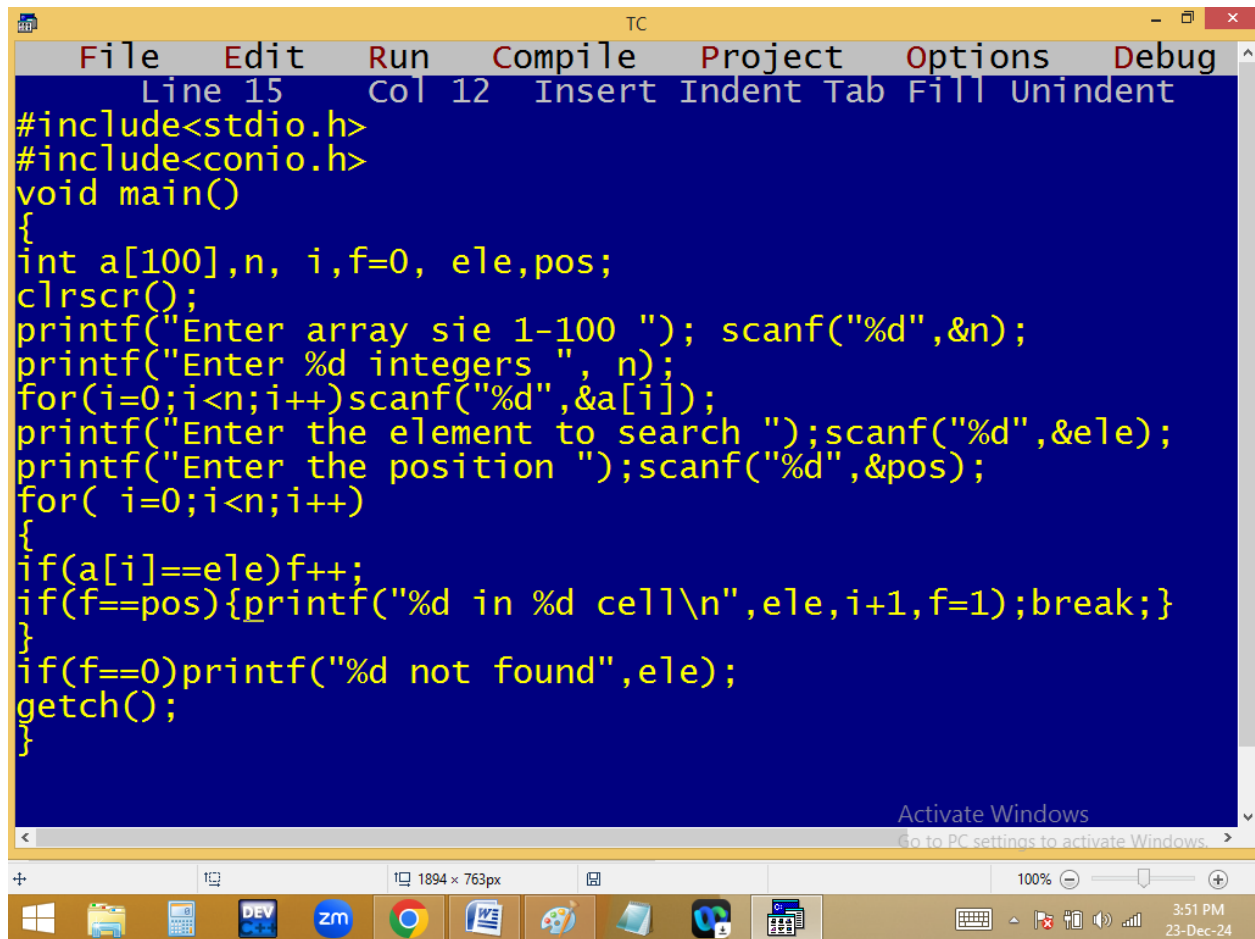
if(f==0)p(ele not found);

0	1	2	3	4
8	5	7	1	20

$\frac{n}{5}$
 $\frac{i}{0}$
 $\frac{ele}{1}$
 $\frac{f}{1}$

1
2
3

Finding nth occurrence of array element:



The image shows a screenshot of a Turbo C++ (TC) IDE window. The title bar reads "TC". The menu bar includes "File", "Edit", "Run", "Compile", "Project", "Options", and "Debug". The status bar at the top indicates "Line 15", "Col 12", and provides options: "Insert", "Indent", "Tab", "Fill", and "Unindent". The main editing area has a dark blue background with yellow text. It contains a C program for searching an element in an array. The code is as follows:

```
#include<stdio.h>
#include<conio.h>
void main()
{
int a[100],n, i,f=0, ele,pos;
clrscr();
printf("Enter array size 1-100 "); scanf("%d",&n);
printf("Enter %d integers ", n);
for(i=0;i<n;i++)scanf("%d",&a[i]);
printf("Enter the element to search ");scanf("%d",&ele);
printf("Enter the position ");scanf("%d",&pos);
for( i=0;i<n;i++)
{
if(a[i]==ele)f++;
if(f==pos){printf("%d in %d cell\n",ele,i+1,f=1);break;}
}
if(f==0)printf("%d not found",ele);
getch();
}
```

At the bottom right of the IDE window, there is a watermark that says "Activate Windows" and "Go to PC settings to activate Windows." The Windows taskbar is visible at the bottom, showing icons for Windows, File Explorer, Calculator, DEV C++, Zoom, Google Chrome, Microsoft Word, Paint, and a folder. The system tray on the right shows the time as 3:51 PM and the date as 23-Dec-24.

